```
100 'NEURAL NETWORK TRAINING (Determination of weights)
110 '
120                                     'INITIALIZE
130 MU = .000005                        'iteration step size
140 DIM X1[101]                         'holds the input layer signal + bias term
150 DIM X2[10]                          'holds the hidden layer signal
160 DIM WH[10,101]                      'holds hidden layer weights
170 DIM WO[10]                          'holds output layer weights
180 '
190 FOR H% = 1 TO 10                    'SET WEIGHTS TO RANDOM VALUES
200   WO[H%] = (RND-0.5)                'output layer weights: -0.5 to 0.5
210   FOR I% = 1 TO 101                 'hidden layer weights: -0.0005 to 0.0005
220     WH[H%,I%] = (RND-0.5)/1000
230   NEXT I%
240 NEXT H%
250 '
260 '                                   'ITERATION LOOP
270 FOR ITER% = 1 TO 800                'loop for 800 iterations
280 '
290   ESUM = 0                          'clear the error accumulator, ESUM
300   '
310   FOR LETTER% = 1 TO 260            'loop for each letter in the training set
320     GOSUB 1000                      'load X1[ ] with training set
330     GOSUB 2000                      'find the error for this letter, ELET
340     ESUM = ESUM + ELET^2            'accumulate error for this iteration
350     GOSUB 3000                      'find the new weights
360   NEXT LETTER%
370   '
380   PRINT  ITER%  ESUM                'print the progress to the video screen
390 '
400 NEXT ITER%
410 '
420 GOSUB XXXX                          'mythical subroutine to save the weights
430 END
```

TABLE 26-2

in the array elements: X2[1] through X2[10]. The variable, X3, contains the network's output value. The weights of the hidden layer are contained in the array, WH[,], where the first index identifies the hidden node (1 to 10), and the second index is the input layer node (1 to 101). The weights of the output layer are held in WO[1] to WO[10]. This makes a total of 1020 weight values that define how the network will operate.

The first action of the program is to set each weight to an arbitrary initial value by using a random number generator. As shown in lines 190 to 240, the hidden layer weights are assigned initial values between -0.0005 and 0.0005, while the output layer weights are between -0.5 and 0.5. These ranges are chosen to be the same order of magnitude that the final weights *must* be. This is based on: (1) the range of values in the input signal, (2) the number of inputs summed at each node, and (3) the range of values over which the sigmoid is active, an input of about $-5 < x < 5$, and an output of 0 to 1. For instance, when 101 inputs with a typical value of 100 are multiplied by the typical weight value of 0.0002, the sum of the products is about 2, which is in the active range of the sigmoid's input.